When Active Learning Does Not Beat Random Labeling for Network Intrusion Detection

Anonymous Workshop Submission

Abstract

Active learning is frequently proposed as a way to reduce the labeling burden in intrusion detection, yet its benefit depends on whether a query strategy selects informative flows rather than difficult or noisy ones. We evaluate random labeling, posterior uncertainty, diversity sampling, and query-by-committee on prepared subsets of UNSW-NB15, CIC-IDS2017, and CSE-CIC-IDS2018. The paper deliberately contrasts balanced random cross-validation with stricter evaluations: repeated-seed paired tests, attack-family and whole-day holdouts, cross-dataset transfer, and a 5% reweighted operating point. On CIC-IDS2017, every non-random strategy is significantly worse than random labeling at a 100-label budget after Bonferroni correction (F1 0.797-0.846 versus 0.889). On separable UNSW-NB15, no alternative remains significantly better than random. The same experiments reveal a broader warning: supervised F1 near 0.97 under balanced random cross-validation falls sharply under day and dataset shift, while recall at a 1% false-positive target is only 0.585-0.668 on the reweighted CIC subset. We do not claim a new detector. Instead, we provide a reproducible negative result: active querying is not a reliable improvement over random labeling on these intrusion benchmarks, and in-distribution evaluation can substantially overstate deployment performance.

1. Introduction

Network intrusion detection must distinguish harmful activity from a large and changing background of legitimate traffic. Labels can make that task much easier, but labeling flows is expensive and often incomplete. Active learning offers an appealing response: ask a human to label only the flows that appear most informative. In security practice, however, queried examples may be ambiguous because of noise, overlap, or collection artifacts rather than because they reduce uncertainty about the deployed decision boundary.

This paper studies a focused empirical question: when a modest label budget is available for flow-based intrusion detection, does active selection beat random labeling? We evaluate a simple, transparent logistic-regression base learner and four selection policies: random sampling, posterior uncertainty, diversity/representativeness sampling, and query-by-committee (QBC). For binary logistic regression, margin and entropy sampling induce the same ranking as posterior uncertainty, so we intentionally report them as one baseline rather than inflate the comparison set.

The central result is negative. On the harder CIC-IDS2017 subset, all three active strategies are significantly worse than random labeling at the fixed budget. On easy UNSW-NB15 data, the strategies are statistically indistinguishable at the performance ceiling. We then ask whether the attractive random-CV results survive stronger tests. They do not consistently survive attack-family, day, cross-dataset, and operating-point shifts. These findings make a limited claim: active querying is not a free improvement for the prepared datasets and protocols studied here.

1.1 Why this comparison is needed

Classical active learning aims to improve a learner with fewer labels by choosing examples that are expected to be especially informative. Uncertainty sampling is the most direct version of this idea: label the point near the model's current boundary. Diversity policies add a coverage intuition by seeking points that represent distinct regions of the pool. Committee methods substitute disagreement among plausible models for the confidence of a single model. These ideas are well established, and each can be sensible when the unlabeled pool resembles the eventual test distribution and labels are consistent enough for boundary refinement to be useful.

Network-flow benchmarks make those assumptions difficult to inspect. Attack labels can group heterogeneous behavior; benign traffic can shift with host, application, collection day, or capture environment; and a detector may face a prevalence far below the balanced research subset. A method that selects boundary cases can therefore spend its scarce labels on examples that are ambiguous in the chosen feature space rather than informative for the target task. Conversely, random labeling may cover common regions of the pool surprisingly well when the model is simple and the benchmark already separates benign and malicious traffic.

Prior intrusion-detection studies commonly report high random-split accuracy or F1, and active-learning papers often compare several query curves on a single benchmark partition. Those studies are valuable baselines, but they do not by themselves answer whether a selection rule gives a statistically reliable benefit at a fixed small budget or whether the resulting detector withstands a collection shift. The present work is structured around those two questions. It deliberately uses transparent models and standard policies so that the negative result cannot be attributed to an opaque architecture or an unmatched implementation.

The objective is not to rule out active learning. A different labeler, feature representation, pool, or cost function can alter the comparison. Instead, the contribution is a falsifiable reference point: under matched budgets and repeated seeds, the commonly used policies tested here do not produce a dependable gain over random selection. This distinction matters when a claimed improvement is intended to save analyst effort rather than merely improve one retrospective learning curve.

Contributions

- A controlled comparison of four genuinely distinct label-acquisition policies under identical budgets, splits, features, and base learner.
- Repeated-seed paired tests showing that each non-random policy is significantly worse than random labeling on CIC-IDS2017, while no policy is significantly better on UNSW-NB15.
- A stricter-evaluation package that separates in-distribution random-CV performance from unseen-family, whole-day, cross-dataset, and reweighted operating-point evidence.
- A reproducible artifact with fixed seeds, metadata-preserving subsets, test coverage, and exact commands for each reported result.

2. Experimental Design

2.1 Data and preprocessing

The study uses three public flow-based benchmark families. Each prepared benchmark subset is balanced for the primary comparisons and retains only numeric traffic features; metadata-retaining variants preserve attack family or day where needed for strict splits. The work does not treat these curated subsets as operational traces. In particular, the 5% experiment below is a reweighted subset, not a naturally observed deployment prevalence.

Dataset	Primary subset	Strict metadata	Purpose
UNSW-NB15	12,000 balanced rows; 21 numeric features	attack family	easy/separable reference; unseen-family tests
CIC-IDS2017	12,000 balanced rows; 78 numeric features	day and attack name	hard active-learning and temporal-shift tests
CSE-CIC-IDS2018	12,000 balanced rows; CICFlowMeter schema	prepared subset	additional benchmark and cross-dataset transfer

Table 1. Prepared data used in the study. All headline limitations are retained in the evaluation and discussion.

2.2 Models and acquisition policies

The unsupervised reference is a vote of z-score, Isolation Forest, Local Outlier Factor, and one-class SVM detectors. The label-budget and active-learning experiments use StandardScaler followed by logistic regression, with class balancing where stated. Each acquisition run begins from a small seed set containing both classes, then requests labels in batches until reaching the specified budget. The same train/test folds and seeds are used for every strategy.

Posterior uncertainty selects points closest to 0.5 under the current classifier. Diversity sampling chooses representative points near clusters in the unlabeled pool. QBC trains a small committee on random 80% subsets of the labeled pool, without replacement, and ranks points by normalized entropy of the members' hard votes. Thus a confident 50/50 committee split is maximally uncertain, as required by the QBC interpretation.

2.3 Evaluation protocol

Primary scores use stratified five-fold cross-validation. The active-learning comparison reports per-seed F1 at a 100-label budget across repeated seeds, then uses paired two-sided Wilcoxon signed-rank tests with Bonferroni correction. This design answers whether observed differences are larger than run-to-run variation, not simply whether one curve appears higher in one split.

We also include stricter tests. Family holdout removes an entire UNSW attack family from supervised training while retaining a benign test split. Day holdout places every CIC flow from a held-out day, benign and attack, in the test set. Cross-dataset transfer trains on the shared CICFlowMeter feature intersection and evaluates on CSE-CIC-IDS2018. Finally, the 5% operating-point experiment down-samples attacks from the prepared balanced subset; the decision threshold is selected in an inner validation split and applied once to the untouched outer test fold.

2.4 Protocol choices that prevent optimistic comparisons

Several implementation choices matter for interpretation. For binary logistic regression, least-confidence, smallest-margin, and entropy rules are monotone transformations of the same posterior uncertainty ranking. Reporting all of them as separate successful methods would count one behavior multiple times, so this study keeps one posterior-uncertainty policy and adds QBC as a substantively different policy. The artifact includes a test that verifies the ranking equivalence for the binary posterior case.

Likewise, QBC disagreement is computed from hard committee votes rather than from the average distance of member probabilities from 0.5. The latter can incorrectly regard a committee split between confident benign and confident attack predictions as certain. Vote entropy instead gives that split the highest disagreement score and gives unanimous votes a score near zero. Committee members train on independently drawn random 80% subsets of the current labeled pool without replacement; this is a lightweight source of model variation, not bootstrap resampling.

The strict day protocol guards against a more subtle form of evaluation optimism. If only attacks from a held-out day are removed while benign examples from the same day remain in training, the test is not a complete day shift. Here the full held-out day is test-only. Similarly, recall at a false-positive target is not computed by choosing the best threshold on the test fold. Each outer fold reserves an inner validation split for threshold choice, and the resulting threshold is applied once to the untouched outer test data.

These controls do not make public benchmarks identical to live networks. They do make the claims easier to audit. A reader can disagree with the choice of data, learner, or metric without needing to infer whether the reported result was helped by a duplicated query strategy, shared temporal rows, or a threshold tuned after observing the evaluation labels.

3. Results

3.1 Labels matter, but the headline can mislead

Table 2 first establishes the familiar result that labels help much more than changing among several standard anomaly detectors. On all three prepared benchmarks, a supervised baseline greatly exceeds the unsupervised vote. This result should not be read as deployment readiness: it is the starting point for the stricter evaluations that follow.

Dataset	Unsupervised ensemble F1	Supervised F1	Difference
UNSW-NB15	0.270	0.996	+0.726
CIC-IDS2017	0.264	0.966	+0.702
CSE-CIC-IDS2018	0.183	0.924	+0.741

Table 2. Balanced random-CV comparisons. These are in-distribution reference results, not deployment estimates.

3.2 Random labeling wins on the hard benchmark

At 100 labels on CIC-IDS2017, random sampling reaches F1 0.889. Posterior uncertainty, diversity, and QBC respectively reach 0.808, 0.797, and 0.846. Each deficit is significant after correction. On UNSW-NB15, all approaches are near the 0.99 ceiling and none survives correction. QBC has the largest UNSW point estimate, but its +0.005 difference is not significant. The result is therefore not that active learning never helps; it is that none of the tested strategies reliably improves upon random labeling under these conditions.

Dataset	Strategy	F1 (95% CI)	Difference vs random	Corrected result
CIC	Random	0.889 [0.882, 0.895]	-	baseline
CIC	Uncertainty	0.808 [0.783, 0.833]	-0.081	worse, p=0.023
CIC	Diversity	0.797 [0.768, 0.827]	-0.091	worse, p=0.023
CIC	QBC	0.846 [0.817, 0.874]	-0.043	worse, p=0.047
UNSW	Random	0.990 [0.987, 0.993]	-	baseline
UNSW	QBC	0.995 [0.995, 0.996]	+0.005	not significant

Table 3. Repeated-seed active-learning statistics at 100 labels. Values are rounded from the frozen results.

3.2.1 Interpreting the negative comparison

The CIC result is stronger than a comparison of two mean scores. Every active policy uses the same label budget, model family, test folds, and repeated seeds as random sampling. The paired analysis therefore tests the acquisition decision itself. The corrected p-values are modest rather than overwhelming, but all three point in the same direction: selecting uncertain, representative, or committee-disputed flows did not improve the classifier and yielded lower F1 than simply labeling randomly selected flows.

This outcome cautions against a common operational shortcut: treating an acquisition heuristic as a guaranteed multiplier on analyst time. The tested policies may concentrate on boundary cases whose labels are inherently variable, on portions of the data distribution that are poorly represented by the fixed feature set, or on examples that improve probability calibration without improving the F1 operating point. The experiments were not designed to distinguish these mechanisms, so we do not attribute the gap to any one cause. Their shared practical implication is enough: the query policy must be evaluated against random labeling under the actual budget and target metric.

3.3 Random-CV success does not transfer consistently

The strict tests put the high supervised random-CV figures in context. In UNSW, holding out one attack family still produces high F1 for most families, although estimates for Backdoor, Worms, and Shellcode are unstable because they contain few attacks. In CIC, entire held-out days behave very differently: Friday and Wednesday retain some separation, while Tuesday and Thursday collapse at the default operating point. Training on CIC shared features and testing on CSE-CIC-IDS2018 reduces F1 to 0.006, despite a 0.516 cross-validation reference on the training dataset's shared features. These are distribution-shift measurements, not evidence of information leakage.

Evaluation	Test setting	Supervised F1	AUC / context
UNSW family holdout	10 held-out families	0.667-0.998	small families widen uncertainty
CIC day holdout	Friday	0.620	AUC 0.897
CIC day holdout	Wednesday	0.750	AUC 0.706
CIC day holdout	Thursday / Tuesday	0.007 / 0.000	AUC 0.674 / 0.688
Cross-dataset	CIC -> CSE-CIC	0.006	AUC 0.574; 27 shared features

Table 4. Generalization tests reveal substantial distribution sensitivity beyond random cross-validation.

3.3.1 What the strict scores do and do not show

The strict results should not be collapsed into a single assertion that the detector has no value. Friday and Wednesday retain nontrivial AUC, and several UNSW family holdouts remain strong. Instead, the evidence shows heterogeneity: a random cross-validation score averages over mixtures that can be much easier than a chronological or cross-collection test. A near-zero F1 at one fixed threshold may coexist with ranking information, but it is still a serious warning for an alerting system whose operating threshold was chosen elsewhere.

The cross-dataset experiment is especially conservative because only a 27-feature shared schema is available for CIC-IDS2017 and CSE-CIC-IDS2018. Its F1 of 0.006 and AUC of 0.574 do not isolate the cause of failure: shifted traffic, preprocessing differences, label construction, and the restricted feature intersection can all contribute. The result nevertheless bounds a tempting interpretation of the training benchmark: a model that scores well under a within-dataset random split should not be presented as cross-environment capable without a direct transfer test.

The family-holdout analysis provides a complementary caution. It is closer to a novelty test because an entire named attack family is removed from training, but its small groups reduce statistical precision. We therefore report it as supporting context rather than as a claim that every unseen attack will be detected. Across all strict evaluations, the stable conclusion is modest: the apparent gap between supervised and unsupervised performance is conditional on the partition and on the target operating point.

3.4 A fixed false-positive target changes the conclusion

A classifier deployed into a security workflow is constrained by alert volume. On a 5% reweighted CIC subset, balanced class weights maximize recall but produce low precision; unweighted logistic regression trades recall for a higher F1. With thresholds selected in inner validation folds, recall at a 1% false-positive target is 0.585 for the weighted model and 0.668 for the unweighted model. The comparable UNSW values are 0.940 and 0.931. Consequently, the balanced CIC F1 of 0.97 is not an adequate summary of the operational tradeoff.

Dataset	Model	Precision	Recall	F1	Recall @ 1% FPR
CIC 5%	balanced weights	0.393	0.946	0.556	0.585
CIC 5%	unweighted	0.926	0.608	0.733	0.668
UNSW 5%	balanced weights	0.868	0.997	0.928	0.940
UNSW 5%	unweighted	0.873	0.975	0.921	0.931

Table 5. Operating-point evaluation on reweighted prepared subsets. Thresholds are chosen without using outer-test labels.

4. Discussion

The active-learning failure is plausible for two nonexclusive reasons. First, a benchmark can be easy enough that random labels rapidly anchor an accurate decision boundary; little room remains for query selection to help. Second, difficult or overlapping flows can be disproportionately selected by uncertainty-based policies. The calibration analyses accompanying this artifact are consistent with the latter explanation on CIC, but the present study does not establish a causal decomposition of noise, overlap, and calibration.

The evaluation also changes how the supervised-versus-unsupervised contrast should be interpreted. Labels provide a large gain within the prepared datasets, but that fact alone does not show that a model will transfer across collection days or benchmarks. The paper's value is therefore methodological: report the random-CV baseline, but pair it with conditions that more closely expose how much of that baseline depends on the data distribution and the thresholding protocol.

4.1 Implications for evaluation and deployment

For researchers, the first implication is comparative rather than prescriptive. A new query strategy should be tested against random labeling with paired repetitions, fixed budgets, and a stated multiplicity correction when several strategies are compared. Reporting only the best seed or an area under a learning curve can obscure whether a practical advantage remains at the label budget an analyst can afford. A negative benchmark result is also useful: it identifies a setting in which a proposed efficiency gain did not materialize and supplies a reproducible baseline for methods that claim to address it.

For practitioners, the second implication is to separate ranking performance from the alerting decision. The day-holdout results retain moderate AUC on some days even when F1 at the default threshold is near zero, while the reweighted evaluation shows a meaningful recall difference at a fixed false-positive target. Neither observation licenses a post-hoc threshold chosen on the test set. Instead, threshold selection must be part of the training and validation procedure, and the chosen metric should reflect the analyst workload that alerts will create.

4.2 Why random can be a strong baseline

Random labeling is sometimes described as a weak baseline because it ignores the current model. At a constrained budget, however, it has two useful properties. It preserves the pool's mixture proportions in expectation and it does not preferentially target a region that the initial model already finds confusing. When the major classes are well represented and the feature space supports a stable linear boundary, those properties can be enough to make the random learner competitive. This is consistent with the UNSW ceiling result, where there is little measurable headroom for any policy.

On CIC, the result is more instructive because there is room to perform worse. All active policies lost to random, including diversity and a genuine vote-disagreement committee. That agreement across policies does not prove that all active learning is harmful, but it weakens explanations that rely on one implementation detail. It also suggests that an evaluation should record not only final F1 but the composition and difficulty of the acquired set. The current artifact preserves the code and seeds needed for such follow-up analysis, while avoiding a causal claim that its measurements do not establish.

A useful next experiment would compare acquisition policies under controlled changes to label noise, class overlap, and prevalence, then relate each policy's selected examples to those changes. Another would attach realistic annotation costs and adjudication rules to labels. Such work could reveal conditions in which active selection recovers an advantage. Until then, the appropriate default is empirical humility: retain random labeling as a serious baseline and require a repeated, statistically tested gain before treating a query strategy as an efficiency improvement.

5. Limitations and Ethics

This is an empirical study of public benchmark subsets, not a deployment study. The CIC day and UNSW family tests contain small groups, so several point estimates have wide unreported uncertainty. The cross-dataset test uses only 27 common CICFlowMeter features, and UNSW has no exact feature-name intersection with CIC. Only one 5% prevalence and one 1% false-positive target are evaluated. The committee is a lightweight random-subsample committee, not an exhaustive comparison against every active-learning method. These limits motivate a narrow conclusion rather than a general claim about all intrusion-detection settings.

All data are public benchmark material and the work is defensive. The artifact contains no live-network scanning or offensive functionality. Any operational adoption would require authorization, privacy review, drift monitoring, and a human process for responding to alerts. The anonymous submission intentionally omits author identity and repository links; the reproducibility package should be shared anonymously only if the target venue permits it.

Several results also depend on choices that were fixed before the final comparisons: the logistic-regression learner, the numeric-feature representation, the batch schedule, and the prepared balanced subsets. A stronger active learner, a representation adapted to encrypted or evolving traffic, or a human-in-the-loop cost model could behave differently. Conversely, the present design intentionally avoids claiming that a small point improvement on a clean benchmark translates into fewer analyst-hours. Future work should evaluate label quality, annotation delay, family novelty, and false-positive cost jointly, ideally on authorized operational data with a documented temporal split.

Finally, the paper uses F1 to make the fixed-budget comparisons legible, even though F1 is not a universal security objective. Different sites may value missed attacks, analyst burden, response latency, or business disruption differently. The operating-point experiment partly addresses this by fixing false-positive rate, but it remains a simplified analysis on a reweighted benchmark subset. Readers should treat the numerical values as reproducible evidence about these protocols, not as a calibrated estimate of a specific organization's risk.

5.1 Interpretation boundaries

The study also does not claim that the reported random baseline is optimal. Random selection could be improved by stratification using information that would be available before labeling, by a changed seed-set design, or by a workflow that treats uncertainty as a review-prioritization signal rather than a label-acquisition rule. Those variants would be separate policies and should be compared using the same fixed-budget, paired protocol. The negative result is specifically about the four policies implemented here, with the specified flow features, learner, batches, and benchmark pools.

Nor should the results be interpreted as a ranking of the public datasets themselves. The apparent difficulty of CIC relative to UNSW in the active-learning comparison may reflect class construction, feature distributions, collection process, or the interaction of those factors with logistic regression. The strict evaluations help reveal that sensitivity but do not identify a ground-truth source. A causal account would require controlled data generation or richer labels describing ambiguity and annotation reliability; neither is available in the current artifact.

These boundaries are deliberate. They turn an otherwise broad claim about active learning into one that can be checked, reproduced, and challenged. A subsequent study can replace the learner, add a cost model, or introduce a new query rule and ask whether it produces a corrected, repeated gain over random sampling under a specified shift. That is a more useful path than interpreting a single benchmark curve as evidence that active learning has solved the labeling problem for intrusion detection.

5.2 Threats to reproducibility

Empirical reproducibility can fail even when the model code is small. Different data-download mirrors, parser versions, random-number defaults, or silent removal of metadata columns can change a benchmark experiment. The companion artifact therefore pins requirements, retains lightweight metadata subsets, records split definitions and seeds, and provides a checklist of exact reproduction commands. Tests cover the query-policy distinctions and the evaluation guards that are most likely to change the study's interpretation.

Reproducibility does not imply that every rerun will yield identical hardware-level timing or that a different library release will report the same floating-point value. The relevant standard here is stronger than a screenshot but narrower than a production audit: an independent reader should be able to reconstruct the comparisons, observe the direction and statistical status of the active-learning result, and identify where a changed dependency or dataset version alters a conclusion. Any public release should archive the precise dataset preparation instructions allowed by the dataset licenses.

The anonymized paper separates these provenance details from identity. Before double-blind submission, the artifact and supplementary material should be checked for author names, local machine paths, account names, revision history, and repository links. That process is also useful outside peer review: provenance that is explicit and machine-readable makes it easier for later users to distinguish a regenerated result from a result that was copied or manually edited.

6. Reproducibility

The companion artifact freezes data-preparation commands, package versions, random seeds, split definitions, and evaluation commands. It includes metadata-preserving subsets, a leakage guard for likely label columns, 66 tests, and scripts for active-learning statistics, strict generalization, cross-dataset transfer, and operating-point evaluation. The artifact is designed so that each table in this paper can be regenerated from a named command. An anonymized repository must remove author identity, machine paths, and application materials before submission.

The artifact checklist distinguishes source-controlled inputs from regenerated outputs. Metadata CSV subsets, scripts, configuration, and tests are versioned; plots, result JSON, and derived tables are intentionally reproducible outputs rather than opaque committed binaries. This separation makes it possible to audit the experiment after changing a dependency while preserving the exact seeds and held-out definitions used for the paper. It also avoids treating a static figure as the only evidence for a numerical claim.

For an archival or double-blind release, the same checklist should be run in a clean environment and its generated outputs compared with the values reported here. Any discrepancy should be described rather than silently normalized away. The paper's claims are narrow enough that this verification is practical: the key question is whether random labeling remains the strongest CIC policy under the frozen protocol and whether the strict evaluations preserve the stated direction of the distribution-shift warning.

7. Conclusion

Active learning is often treated as an automatic answer to limited labels in security telemetry. Across the prepared benchmarks studied here, it is not. Random labeling matches all alternatives on separable UNSW-NB15 data and significantly outperforms posterior uncertainty, diversity, and QBC on CIC-IDS2017. Stricter day, cross-dataset, and operating-point evaluations further show that impressive balanced random-CV scores can be poor guides to distribution-shift performance. The practical lesson is modest but useful: evaluate the acquisition policy, the split, and the operating point together before claiming that active learning reduces the real labeling burden of intrusion detection.

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